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High-mobility-electron transistors having heat dissipating structures

Abstract

A semiconductor device is provided. The semiconductor device includes a substrate, a semiconductor layer, a device layer, and heat dissipating structures. The semiconductor layer is over the substrate and the device layer is over the semiconductor layer. The device layer includes a first ohmic contact and a second ohmic contact. The heat dissipating structures are at least through the substrate and the semiconductor layer, and between the first ohmic contact and the second ohmic contact.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6744072	12/2003	Romano	257/85	H01S 5/02461
8575657	12/2012	Gambin	257/706	H01L 23/3732

OTHER PUBLICATIONS

Srikant Kumar Mohanty et al., Thermal Management of GaN-on-Si High Electron Mobility Transistor by Copper Filled Micro-Trench Structure, Scientific Reports, 2019, pp. 1-9, vol. 9—Issue No. 19691, The Authors. cited by applicant

Daniel Shoemaker et al., Diamond-Incorporated Flip-Chip Integration for Thermal Management of GaN and Ultra-Wide Bandgap RF Power Amplifiers, IEEE Transactions on Components, Packaging and Manufacturing Technology, Aug. 2021, pp. 1177-1186, vol. 11—Issue No. 8, IEEE. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to semiconductor devices, and more particularly to high-electron-mobility transistors (HEMTs) with heat dissipating structures and methods of forming the same.

BACKGROUND

(2) Semiconductor devices, such as high-electron-mobility transistors (HEMTs), have a desirable intrinsic capacity for handling large current and power densities due to the presence of a junction between two semiconductor materials with different band gaps. HEMTs are able to operate at higher frequencies and higher temperatures than the typical field-effect transistors (FETs), for example, metal-oxide-semiconductor FETs (MOSFETs), and are often used in high-frequency products, for example, cell phones, satellite television receivers, voltage converters, and radar equipment.

(3) One of the main limitations of using HEMTs is the self-heating effect during operation, which degrades electrical performance and causes reliability problems. For example, when the temperature of the HEMT exceeds a threshold temperature, there is a reduction in carrier mobility, as well as the average lifespan thereof.

(4) Therefore, solutions are provided to overcome, or at least ameliorate, the disadvantages

described above.

SUMMARY

(5) To achieve the foregoing and other aspects of the present disclosure, high-electron-mobility transistors (HEMTs) with heat dissipation structures and methods of forming the same are presented.

(6) According to an aspect of the present disclosure, a semiconductor device is provided. The semiconductor device includes a substrate, a semiconductor layer, a device layer, and a heat dissipating structure. The semiconductor layer is over the substrate and the device layer is over the semiconductor layer. The device layer includes a first ohmic contact and a second ohmic contact. The heat dissipating structures are at least through the substrate and the semiconductor layer, and between the first ohmic contact and the second ohmic contact.

(7) According to another aspect of the present disclosure, a semiconductor device is provided. The semiconductor device includes a substrate, a first semiconductor layer, a second semiconductor layer, a device layer, a through-substrate via (TSV) structure, and heat dissipating structures. The first semiconductor layer is over the substrate, the second layer semiconductor is over the first semiconductor layer, and the device layer is over the second semiconductor layer. The device layer includes a first ohmic contact and a second ohmic contact. The TSV structure is through the substrate, the first semiconductor layer, and the second semiconductor layer, and directly under the first ohmic contact. The heat dissipating structures are at least through the substrate and the first semiconductor layer, and between the first ohmic contact and the second ohmic contact.

(8) According to yet another aspect of the present disclosure, a method of forming a semiconductor device is provided. The method includes forming a semiconductor layer over a substrate and forming a device layer over the semiconductor layer. The device layer includes a first ohmic contact and a second ohmic contact spaced apart from the first ohmic contact by a distance. A through-substrate via (TSV) structure and heat dissipating structures are formed at least through the substrate and the semiconductor layer. The TSV structure is electrically coupled to the first ohmic contact and the heat dissipating structures are arranged between the first ohmic contact and the second ohmic contact.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments of the present disclosure will be better understood from a reading of the following detailed description, taken in conjunction with the accompanying drawings:

(2) FIG. 1 illustrates a cross-sectional view of a transistor, according to an embodiment of the disclosure.

(3) FIG. 2 illustrates an alternative embodiment of the transistor, according to an embodiment of the disclosure.

(4) FIGS. 3A to 3C are cross-sectional views of a transistor, illustrating a method of forming heat dissipating structures therein, according to an embodiment of the disclosure.

(5) For simplicity and clarity of illustration, the drawings illustrate the general manner of construction, and certain descriptions and details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the discussion of the described embodiments of the disclosure.

(6) Additionally, features in the drawings are not necessarily drawn to scale. For example, the dimensions of some of the features in the drawings may be exaggerated relative to other features to help improve understanding of embodiments of the device. The same reference numerals in different drawings denote the same features, while similar reference numerals may, but do not necessarily, denote similar features.

DETAILED DESCRIPTION

(7) The present disclosure relates generally to high-electron-mobility transistors (HEMTs) with heat dissipation structures and methods of forming the same. Various embodiments of the present disclosure are now described in detail with accompanying drawings. It is noted that like and corresponding features are referred to by the use of the same reference numerals. The embodiments disclosed herein are exemplary, and not intended to be exhaustive or limiting to the disclosure.

(8) FIG. 1 illustrates a cross-sectional view of a transistor **100**, according to an embodiment of the disclosure. FIG. 2 illustrates a cross-sectional view of the transistor **100**, according to an alternative embodiment of the disclosure. The transistor **100** may function as a high-electron-mobility transistor (HEMT).

(9) The transistor **100** may include a substrate **102** having a substrate surface **104**. The substrate **102** may be a bulk substrate as illustrated in the accompanying drawings. Alternatively, the substrate **102** may be a composite substrate, for example, a semiconductor-on-insulator substrate or a semiconductor-on-ceramic substrate.

(10) The substrate **102** may include a semiconductor material, for example, silicon, germanium, silicon germanium, silicon carbide, or other semiconductor compounds, such as group II-VI semiconductor compounds including zinc selenide, or group III-V semiconductor compounds, including gallium arsenide or gallium nitride. Alternatively, the substrate **102** may include sapphire, spinel, zinc oxide, aluminum nitride, or any other material capable of supporting the fabrication of a semiconductor material thereupon.

(11) The transistor **100** may further include a device layer **106**, a barrier layer **108**, a channel layer **110**, and a buffer layer **112** over the substrate **102**. The device layer **106** may include various conductive features, for example, a gate structure **114** and interconnect structures (not shown), and various insulative features, for example, dielectric layers (not shown) and spacers (not shown).

(12) The device layer **106** may further include an ohmic contact **116** and an ohmic contact **118** spaced apart from the ohmic contact. In an embodiment of the disclosure, the ohmic contacts **116**, **118** may be arranged over the barrier layer **108**, as illustrated in FIG. 1. The ohmic contacts **116**, **118** may include bottom surfaces **120** in direct contact with the barrier layer **108**. In another embodiment of the disclosure, the ohmic contacts **116**, **118** may be arranged at least partially within the barrier layer **108**. For example, the ohmic contacts **116**, **118** may penetrate through the barrier layer **108**, as illustrated in FIG. 2. The bottom surfaces **120** of the ohmic contacts **116**, **118** may then be in direct contact with the channel layer **110**. In another example, the ohmic contacts **116**, **118** may be arranged partially within the barrier layer **108** such that a portion of the barrier layer **108** may be arranged under the ohmic contacts **116**, **118**, and the bottom surfaces **120** of the ohmic contacts **116**, **118** may terminate within the barrier layer **108**.

(13) The ohmic contacts **116**, **118** may function as either a source terminal or a drain terminal of the transistor **100**, and the gate structure **114** may be arranged between the ohmic contacts **116**, **118** in the device layer **106**. In an embodiment of the disclosure, the ohmic contact **116** may function as a source terminal and the ohmic contact **118** may function as a drain terminal. Alternatively, the ohmic contact **116** may function as a drain terminal and the ohmic contact **118** may function as a source terminal.

(14) The barrier layer **108** may be arranged under the device layer **106**. The barrier layer **108** may include one or more layers of semiconductor material, for example, a group III-V semiconductor compound, including aluminum gallium arsenide or aluminum gallium nitride.

(15) The channel layer **110** may be arranged under barrier layer **108**. The channel layer **110** may include one or more layers of a semiconductor material having a lower band gap than that of the barrier layer **108**. For example, the channel layer **110** may include a group III-V semiconductor compound, such as gallium arsenide, indium gallium arsenide, or gallium nitride. In an embodiment of the disclosure, the channel layer **110** may include the same semiconductor material as the substrate **102**. Alternatively, the channel layer **110** may include a semiconductor material that

is different from the semiconductor material of the substrate **102**.

(16) The interface between the barrier layer **108** and the channel layer **110** may provide a heterojunction, that results in the formation and confinement of a two-dimensional electron gas (2DEG) (not shown) at the interface. The 2DEG may be induced by polarization mismatches between the barrier layer **108** and the channel layer **110**. The 2DEG may provide a conductive pathway between the ohmic contacts **116**, **118** with increased carrier mobility. In an embodiment of the disclosure, the portion of the 2DEG between the ohmic contacts **116**, **118** defines a channel region of the transistor **100**, and the distance between the ohmic contacts **116**, **118** may be referred to as the effective channel length C of the transistor **100**.

(17) The buffer layer **112** may be arranged between the substrate **102** and the channel layer **110**. The buffer layer **112** may serve to accommodate any lattice mismatch between the substrate **102** and the channel layer **110**, thereby improving the “on” state and “off” state electrical characteristics of the transistor **100**. The buffer layer **112** may also serve to accommodate any strain and thermal expansion coefficients mismatch between the substrate **102** and the channel layer **110**, thereby minimizing potential deformation of the substrate **102**, such as bowing, warping, cracking, or breaking thereof. In an embodiment of the disclosure, the buffer layer **112** and the channel layer **110** may have substantially equal thicknesses. In another embodiment of the disclosure, the buffer layer **112** and the channel layer **110** may have a total thickness of about 2 micrometers (m) to about 5 μm .

(18) The buffer layer **112** may include a semiconductor material that is different from the channel layer **110**. For example, the buffer layer **112** may include a group III-V semiconductor compound, including aluminum nitride, aluminum gallium nitride, or gallium nitride. In an embodiment of the disclosure, the buffer layer **112** may include a single-layered structure. Alternatively, the buffer layer **112** may include a superlattice structure with multiple sublayers of group III-V semiconductor compounds, for example, alternating layers of aluminum nitride and gallium nitride.

(19) The transistor **100** may further include a through-substrate via (TSV) structure **122** arranged through the substrate **102**, the buffer layer **112**, the channel layer **110**, and the barrier layer **108**. The TSV structure **122** may be arranged directly under and electrically coupled to the ohmic contact **116**. The TSV structure **122** may serve as a vertical interconnection for the transistor **100**, enabling the transistor **100** to be vertically stacked above and electrically coupled to another conductive feature, for example, a semiconductor device or a package to create three-dimensional integrated circuits or three-dimensional packages. The TSV structure **122** may include an electrically conductive material, for example, copper, gold, tungsten, tantalum, hafnium, titanium, silver, cobalt, or combinations thereof.

(20) The TSV structure **122** may include a TSV portion **124** arranged through the barrier layer **108** and at least partially through the channel layer **110**, and another TSV portion **128** arranged through the substrate **102** and at least partially through buffer layer **112**; the transition of the TSV portion **124** to the TSV portion **128** is diagrammatically shown with a dash-dot-dot line for purposes of illustration. The TSV structure **122** may have a width dimension that varies along the length of its height H1. For example, the TSV portions **124**, **128** may each taper (i.e., narrows) in a direction towards the device layer **106**. In another example, the width dimensions of the TSV portions **124**, **128** may increase with increasing distance from the device layer **106**. In an embodiment of the disclosure, the sidewall **126** of the TSV portion **124** and the sidewall **130** of the TSV portion **128** may be substantially planar, though not necessarily coplanar with each other. In another embodiment of the disclosure, the width dimension of the TSV portion **124** and/or the width dimension of the TSV portion **128** may vary based on a linear function to provide a trapezoidal shape, though not necessarily the same linear function. In yet another embodiment of the disclosure, the width dimension of the TSV portion **124** and/or the width dimension of the TSV portion **128** may vary based on a non-linear function.

(21) Alternatively, the TSV portions **124**, **128** may each include substantially vertically straight

sidewalls. Accordingly, the TSV portions **124**, **128** may each have a substantially constant width dimension, and the width dimension of the TSV portion **124** may be narrower than the width dimension of the TSV portion **128**.

(22) The TSV structure **122** may include a step surface **132** that may be substantially parallel to the substrate surface **104**. The step surface **132** may transition the TSV portion **124** to the TSV portion **128**. For example, the TSV portion **124** may have a width $W1.sub.A$ and the TSV portion **128** may have a width $W1.sub.B$ at a plane substantially coplanar with the step surface **132**, and the width $W1.sub.A$ of the TSV portion **124** is narrower than the $W1.sub.B$ of the TSV portion **128**. In another example, the sidewall **126** of the TSV portion **124** may be arranged at an inward offset from the sidewall **130** of the TSV portion **128**, and the step surface **132** may connect the sidewall **130** of the TSV portion **128** and the sidewall **126** of the TSV portion **124**.

(23) The step surface **132** of the TSV structure **122** may be arranged in the buffer layer **112** such that the TSV portion **124** may have a height $H1.sub.A$ extending from the bottom surface **120** of the ohmic contact **116** to a level within the buffer layer **112**. For example, the TSV portion **124** may terminate at a level below the interface between the channel layer **110** and the buffer layer **112**, as illustrated in FIG. 1. The TSV portion **128** may have a height $H1.sub.B$ extending from the TSV portion **124** (or the step surface **132**) to the substrate surface **104**.

(24) Alternatively, as illustrated in FIG. 2, the step surface **132** of the TSV structure **122** may be arranged in the channel layer **110** such that the height $H1.sub.A$ of the TSV portion **124** may extend from the bottom surface **120** of the ohmic contact **116** to a level within the channel layer **110**. For example, the TSV portion **124** may terminate at a level above the interface between the channel layer **110** and the buffer layer **112**. The height $H1.sub.B$ of the TSV portion **128** may extend from the TSV portion **124** (or the step surface **132**) to the substrate surface **104**. In an embodiment of the disclosure, the height $H1.sub.A$ of the TSV portion **124** is shorter than the height $H1.sub.B$ of the TSV portion **128**.

(25) Even though FIG. 1 illustrates two TSV structures **122** being electrically coupled to the ohmic contact **116**, it is understood that the number of TSV structures **122** may vary according to the design of the transistor, for example, a transistor may include one TSV structure **122** or more than two TSV structures **122**.

(26) The transistor **100** may yet further include heat dissipating structures **134**, such as a bulk conductive structure **136** and a conductive liner **138**. The heat dissipating structures **134** may serve as a passive heat spreader to dissipate heat from the transistor **100** when heat tends to be generated. The heat dissipating structures **134** may be arranged at least through the substrate **102** and the buffer layer **112**. For example, a portion of the bulk conductive structure **136** and the conductive liner **138** may be in direct contact with the channel layer **110**, as illustrated in FIG. 1. In another example, the bulk conductive structure **136** and a conductive liner **138** may be arranged through the substrate **102** and the buffer layer **112** and partially within the channel layer **110**.

(27) During the operation of the transistor **100**, the ohmic contacts **116**, **118** may be subjected to high bias conditions which may result in the generation of a large amount of heat in the channel layer **110**, particularly in the 2DEG under the gate structure **114**. As the buffer layer **112** may include materials having poor heat conductance, the large amount of heat generated may be trapped within the channel layer **110** and inadvertently causes the temperature in the channel layer **110** to rise. When the heat is not being efficiently and sufficiently removed therefrom, the transistor **100** may suffer from substantial degradation of electrical performance and reduction in reliability. By arranging the heat dissipating structures **134** in proximity to the region where heat tends to be generated, for example, the channel region, the heat accumulated during the operation of the transistor **100** may be effectively transferred to the heat dissipating structures **134** to be removed therefrom.

(28) The bulk conductive structure **136** may be arranged at least through the substrate **102** and may be further arranged under and between the ohmic contacts **116**, **118**. The bulk conductive structure

136 may have a width **W2** at the top surface **142** thereof. In an embodiment of the disclosure, the width **W2** of the bulk conductive structure **136** may be at least as wide as the effective channel length **C** of the transistor **100**. For example, as illustrated in FIGS. **1** and **2**, the width **W2** of the bulk conductive structure **136** may be wider than the effective channel length **C** of the transistor **100**, and the side portions of the bulk conductive structure **136** may be arranged under the ohmic contacts **116**, **118**.

(29) Alternatively, in another embodiment of the disclosure, the width **W2** of the bulk conductive structure **136** may be narrower than the effective channel length **C** of the transistor **100**, as illustrated in FIG. **2**. The width **W2** of the bulk conductive structure **136** may be significantly wider than the widths **W1.sub.A**, **W1.sub.B** of the TSV structure **122** to provide sufficient thermal capacity to effectively function as a heat dissipating structure. In an embodiment of the disclosure, the width **W2** of the bulk conductive structure **136** may be no wider than 10 μm .

(30) The width dimension of the bulk conductive structure **136** may vary along the length of its height **H2**. For example, the bulk conductive structure **136** may taper (i.e., narrows) in a direction towards the device layer **106**, and the width **W2** of the bulk conductive structure **136** may be the minimum width thereof. In another example, the width dimensions of the bulk conductive structure **136** may increase with increasing distance from the device layer **106**. In an embodiment of the disclosure, the sidewalls of the bulk conductive structure **136** may be substantially planar. In another embodiment of the disclosure, the width dimension of the bulk conductive structure **136** may vary based on a linear function to provide a trapezoidal shape. In yet another embodiment of the disclosure, the width dimension of the bulk conductive structure **136** may vary based on a non-linear function. Alternatively, the bulk conductive structure **136** may include substantially vertically straight sidewalls and may have a substantially constant width dimension along the length of its height.

(31) The height **H2** of the bulk conductive structure **136** may be substantially equal to the height **H1.sub.B** of the TSV structure **122**. For example, the height **H2** of the bulk conductive structure **136** may extend from the substrate surface **104** to a plane substantially coplanar with the step surface **132** of the TSV structure **122**. As illustrated in FIG. **1**, the top surface **142** of the bulk conductive structure **136** may terminate within the buffer layer **112**. Alternatively, as illustrated in FIG. **2**, the top surface **142** of the bulk conductive structure **136** may terminate within the channel layer **110**. In an embodiment of the disclosure, the top surface **142** of the bulk conductive structure **136** may be substantially coplanar with the step surface **132** of the TSV structure **122**.

(32) The bulk conductive structure **136** may include an electrically conductive material, preferably one having a high thermal conductivity. In an embodiment of the disclosure, the bulk conductive structure **136** may include the same electrically conductive material as the TSV structure **122**, as illustrated in FIGS. **1** and **2**. In another embodiment of the disclosure, the bulk conductive structure **136** may include an electrically conductive material that is different from the TSV structure **122**.

(33) The conductive liner **138** may be arranged at least through the substrate **102**. The conductive liner **138** may overlie the sidewalls **140** and the top surface **142** of the bulk conductive structure **136**. The conductive liner **138** may be further arranged partially on the sidewalls of the TSV structure **122**. For example, the conductive liner **138** may be arranged on the sidewalls of the TSV portion **128** and may overlie the step surface **132**.

(34) Additionally, the conductive liner **138** may overlie the substrate surface **104**. The conductive liner **138** may have a substantially equal thickness. In an embodiment of the disclosure, the conductive liner **138** may have a thickness of no more than 0.1 μm .

(35) The conductive liner **138** may effectively assist the bulk conductive structure **136** and the TSV structure **122** in dissipating heat from the transistor **100** while providing electrical isolation between adjacent conductive features, for example between the TSV structure **122** and the bulk conductive structure **136**.

(36) The conductive liner **138** may include a thermally conductive yet electrically insulative

material, for example, a carbon-based material. Examples of a carbon-based material may include an elemental carbon material, such as diamond, or a carbon-based compound such as silicon carbide. As used herein, the term “elemental carbon material” includes not only a single-element carbon but may also include other trace elements in various alloyed combinations with carbon, as long as the chemical and physical properties of element carbon remain substantially unaffected. The conductive liner **138** may also include other thermally conductive yet electrically insulative materials, for example, aluminum oxide, aluminum nitride, boron nitride, or beryllium oxide.

(37) FIGS. 3A to 3C are cross-sectional views of a transistor **200**, illustrating a method of forming heat dissipating structures therein, according to an embodiment of the disclosure. The transistor **200**, in contrast to the transistor **100** in FIG. 1, may be arranged in a flipped configuration for the ease of fabrication of the heat dissipating structures. For example, the transistor **200** may be inverted such that the substrate surface **104** may be over the device layer **106**. The transistor **200** may be similar to the transistor **100** in FIG. 1, and thus common features are labeled with the same reference numerals and need not be discussed. Certain structures may be conventionally fabricated, for example, using known processes and techniques, and specifically disclosed processes and methods may be used to achieve individual aspects of the present disclosure.

(38) FIG. 3A illustrates the transistor **200** after forming openings **244**, **246** at least through the substrate **102** and the buffer layer **112**, according to an embodiment of the disclosure. The transistor **200** may have completed the fabrication of the device layer **106**, including various conductive features, for example, a gate structure **114**, interconnect structures, and ohmic contacts **116**, **118**, and various insulative features, for example, dielectric layers and spacers.

(39) The substrate **102** may undergo a backside grinding process to thin down its thickness before forming the openings **244**, **246**. In an embodiment of the disclosure, the substrate **102** may have a thickness of about 50 μm to about 100 μm after the backside grinding process.

(40) The openings **244**, **246** may at least extend through the substrate **102** and the buffer layer **112** such that a portion of the channel layer **110** may be exposed in the openings **244**, **246**. For example, to form the transistor **100** in FIG. 1, the openings **244**, **246** may extend from the substrate surface **104** and terminate at the interface between the channel layer **110** and the buffer layer **112**. In another example, to form the transistor **100** in FIG. 2, the openings **244**, **246** may extend from the substrate surface **104** and terminate within the channel layer **110**.

(41) The openings **244**, **246** may be formed using a patterning technique, including lithography and etching processes. In an embodiment of the disclosure, the opening **244** may be formed concurrently with the opening **246**. In another embodiment of the disclosure, the opening **244** may be formed in a separate fabrication step as the opening **246**.

(42) The openings **244**, **246** may be arranged over the ohmic contacts **116**, **118**. For example, the opening **244** may be arranged directly above the ohmic contact **116** and the opening **246** may be arranged between the ohmic contacts **116**, **118**. The opening **244** may have a width $W3$ at the bottom surface thereof. The opening **246** may have a width $W4$ at the bottom surface thereof and the width $W4$ at least as wide as the effective channel length C of the transistor **200**. For example, the width $W4$ of the opening **246** may be wider than the effective channel length C , and end portions of the ohmic contacts may be arranged directly under the opening **246**. The width $W4$ of the opening **246** may be significantly wider than the width $W3$ of the opening **244**. In an embodiment of the disclosure, the opening **244** may be an opening for a part of a TSV structure of the transistor **200**. In another embodiment of the disclosure, the opening **246** may be an opening for a heat dissipating structure of the transistor **200**.

(43) FIG. 3B illustrates the transistor **200** after forming a conductive liner **138**, according to an embodiment of the disclosure. The conductive liner **138** may be conformally deposited in the openings **244**, **246**, for example, the conductive liner **138** may be deposited on the sidewalls and bottom surfaces thereof. The conductive liner **138** may be further deposited on the substrate surface **104**. As a result of the deposition of the heat dissipating liner **160** in the openings **244**, **246**, the

effective width **W3** of the opening **244** in FIG. 3A may be reduced to a width **W5** that may be similar to the width **W1.sub.B** of the TSV portion **128** in FIG. 1. Similarly, the width **W4** of the opening **246** in FIG. 3A may be reduced to a width **W6** that may be similar to the width **W2** of the bulk conductive structure **136** in FIG. 1. In an embodiment of the disclosure, the conductive liner **138** may have a thickness of no more than 0.1 μm .

(44) The conductive liner **138** may include a thermally conductive yet electrically insulative material, for example, an elemental carbon material, such as diamond, or a carbon-based compound such as silicon carbide, aluminum oxide, aluminum nitride, boron nitride, or beryllium oxide. The conductive liner **138** may be deposited using a conformal deposition technique, including a chemical vapor deposition (CVD) process or an atomic layer deposition (ALD) process.

(45) FIG. 3C illustrates the transistor **200** after forming a TSV opening **248**, according to an embodiment of the disclosure. An opening **250** may be formed directly under the opening **244**, extending through the channel layer **110** and the barrier layer **108**; the transition of the opening **244** to the opening **250** is diagrammatically shown with a dash-dot-dot line for purposes of illustration. The opening **250** may expose a portion of the ohmic contact **116** therewithin. The opening **250** may be formed using a patterning technique, including lithography and etching processes. The openings **244**, **250** may form the TSV opening **248** of the transistor **200**. In an embodiment of the disclosure, the sidewalls of the openings **244**, **250** may be substantially planar, though not necessarily coplanar with each other.

(46) Additional fabrication steps may be performed before forming the opening **250**, for example, forming one or more protection layers over the opening **246** that will be not subjected to the patterning technique.

(47) The opening **250** may have a narrower width than the opening **244**. For example, the TSV opening **248** may include a step surface **132** that transitions the opening **244** to the opening **250** such that the sidewalls of the opening **250** may be arranged at an inward offset from the sidewalls of the opening **244**.

(48) Processing continues with the deposition of one or more liner films (not shown) and the deposition of an electrically conductive material to at least fill the opening **246** and the TSV opening **248**. The TSV opening **248** may be filled to form the TSV structure **122** and opening **246** may be filled to form the bulk conductive structure **136** in FIG. 1. The liner film may be deposited using a conformal deposition technique, including a CVD process, a physical vapor deposition (PVD) process, or an ALD process. The electrically conductive material may be deposited over the liner film using a deposition technique, including electroplating. The electrically conductive material may be deposited over the substrate surface **104** and may serve as a redistribution layer (RDL).

(49) The one or more liner films may include barrier liner and/or adhesion liner, including titanium, tantalum, tantalum nitride, tantalum silicon nitride, tungsten, titanium nitride, or titanium silicon nitride. The electrically conductive material may include copper, tungsten, tantalum, hafnium, titanium, silver, cobalt, or combinations thereof.

(50) The terms “top”, “bottom”, “over”, “under”, and the like in the description and the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the devices described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

(51) Additionally, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the first and second features, such that the first and second features may not be in direct contact.

(52) Similarly, if a method is described herein as involving a series of steps, the order of such steps as presented herein is not necessarily the only order in which such steps may be performed, and

certain of the stated steps may possibly be omitted and/or certain other steps not described herein may possibly be added to the method. Furthermore, the terms “comprise”, “include”, “have”, and any variations thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or device that comprises a list of features is not necessarily limited to those features but may include other features not expressly listed or inherent to such process, method, article, or device. Occurrences of the phrase “in an embodiment” herein do not necessarily all refer to the same embodiment.

(53) In addition, unless otherwise indicated, all numbers expressing quantities, ratios, and numerical properties of materials, reaction conditions, and so forth used in the specification and claims are to be understood as being modified in all instances by the term “about”.

(54) Furthermore, approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related.

(55) Accordingly, a value modified by a term or terms, such as “approximately”, “about”, “substantially” is not limited to the precise value specified. In some instances, the approximating language may correspond to the precision of an instrument for measuring the value. In other instances, the approximating language may correspond to within normal tolerances of the semiconductor industry. For example, “substantially coplanar” means substantially in a same plane within normal tolerances of the semiconductor industry, and “substantially perpendicular” means at an angle of 90 degrees plus or minus a normal tolerance of the semiconductor industry.

(56) While several exemplary embodiments have been presented in the above detailed description of the device, it should be appreciated that a number of variations exist. It should further be appreciated that the embodiments are only examples, and are not intended to limit the scope, applicability, dimensions, or configuration of the device in any way.

(57) Rather, the above detailed description will provide those skilled in the art with a convenient road map for implementing an exemplary embodiment of the device, it being understood that various changes may be made in the function and arrangement of features and methods of fabrication described in an exemplary embodiment without departing from the scope of this disclosure as set forth in the appended claims.

Claims

1. A semiconductor device, comprising: a substrate; a semiconductor buffer layer over the substrate; a device layer over the semiconductor buffer layer, the device layer comprising a first ohmic contact and a second ohmic contact; and heat dissipating structures at least through the substrate and the semiconductor buffer layer, and between the first ohmic contact and the second ohmic contact.
2. The semiconductor device of claim 1, wherein the heat dissipating structures comprises: a bulk conductive structure; and a conductive liner conformally overlying the bulk conductive structure.
3. The semiconductor device of claim 2, wherein the first ohmic contact is spaced apart from the second ohmic contact by a distance, and the bulk conductive structure has a width that is at least as wide as the distance between the first ohmic contact and the second ohmic contact.
4. The semiconductor device of claim 2, wherein the first ohmic contact is spaced apart from the second ohmic contact by a distance, and the bulk conductive structure has a width that is narrower than the distance between the first ohmic contact and the second ohmic contact.
5. The semiconductor device of claim 2, wherein the conductive liner comprises a thermally conductive and electrically insulative material.
6. The semiconductor device of claim 5, wherein the thermally conductive and electrically insulative material comprises a carbon-based material.
7. The semiconductor device of claim 2, wherein the bulk conductive structure comprises an

electrically and thermally conductive material.

8. A semiconductor device, comprising: a substrate; a semiconductor buffer layer over the substrate; a semiconductor channel layer over the semiconductor buffer layer; a device layer over the semiconductor channel layer, the device layer comprising a first ohmic contact and a second ohmic contact; a through-substrate via (TSV) structure through the substrate, the semiconductor buffer layer, and the semiconductor channel layer, the TSV structure is directly under the first ohmic contact; and heat dissipating structures at least through the substrate and the first semiconductor buffer layer, the heat dissipating structures are between the first ohmic contact and the second ohmic contact.

9. The semiconductor device of claim 8, wherein the TSV structure further comprising: a lower TSV portion at least through the substrate and the semiconductor buffer layer; an upper TSV portion over the lower TSV portion, the upper TSV portion is at least partially through the semiconductor channel layer and electrically coupled to the first ohmic contact; and a step surface between the upper TSV portion and the lower TSV portion.

10. The semiconductor device of claim 9, wherein the lower TSV portion comprises a sidewall, the upper TSV portion comprises a sidewall, and the sidewall of the upper TSV portion is at an inward offset from the sidewall of the lower TSV portion.

11. The semiconductor device of claim 9, wherein the heat dissipating structures comprises: a bulk conductive structure; and a conductive liner, the conductive liner conformally overlying the bulk conductive structure and the lower TSV portion.

12. The semiconductor device of claim 11, wherein the bulk conductive structure has a top surface, and the top surface of the bulk conductive structure is substantially coplanar with the step surface of the TSV structure.

13. The semiconductor device of claim 11, wherein the bulk conductive structure comprises a conductive material and the TSV structure comprises a conductive material that is the same conductive material as the bulk conductive structure.

14. The semiconductor device of claim 8, wherein the first semiconductor buffer layer and the semiconductor channel layer comprise different group III-V semiconductor compounds.
